

Periodicities in Sunspot Activity and Volcanic Eruptions: A Wavelet Analysis



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Abstract:

Volcanic eruptions are driven by a complex interplay of internal geologic processes including but not limited to magma supply, tectonic stress, crustal structure, and external ("exogenic") influences beyond the solid Earth. Yet the role of solar variability in modulating eruption frequency remains under-explored. Understanding correlations between solar and volcanic activity could provide further insights into the factors that contribute to an individual volcano's likelihood of eruption and help improve prediction models. This research investigates and compares the periodicities of average annual sunspot numbers and annual global total volcanic eruptions over time (from 1700 to 2024). Sunspot data was obtained from the Royal Observatory of Belgium's Sunspot Index and Long-term Solar Observations, and the data on volcanic eruptions came from the National Oceanic and Atmospheric Administration.

Using RStudio, a KZ filter was applied to remove noise while still keeping the first and last few years. Determining the most prominent periodicities (also in RStudio) was achieved by conducting a wavelet analysis and locating the local peaks. Based on these peaks, average annual sunspot numbers and the annual total number of eruptions show some similar cycles, notably in the 2-year, 21-year, and 168 bands. Although nothing has been definitively proven, these correlations indicate that solar activity may subtly and indirectly alter the probability of a single volcano erupting, possibly due to the sun's effects on the atmosphere or other terrestrial occurrences. Understanding this influence allows for a deeper understanding of the science behind an eruption event, helping people more accurately predict when a volcano might erupt and consequently minimize the damage.

Background Information:

- Sunspots are a result of the sun's magnetic field and are therefore a measurement of solar activity (figure 1).
- Around 800 million people live near an active volcano (pubs.usgs.gov).
- Volcanic eruptions affect people all around the globe (figure 2).
- Previous research has been done on the correlation between volcanic and solar activity (Stothers 1989).

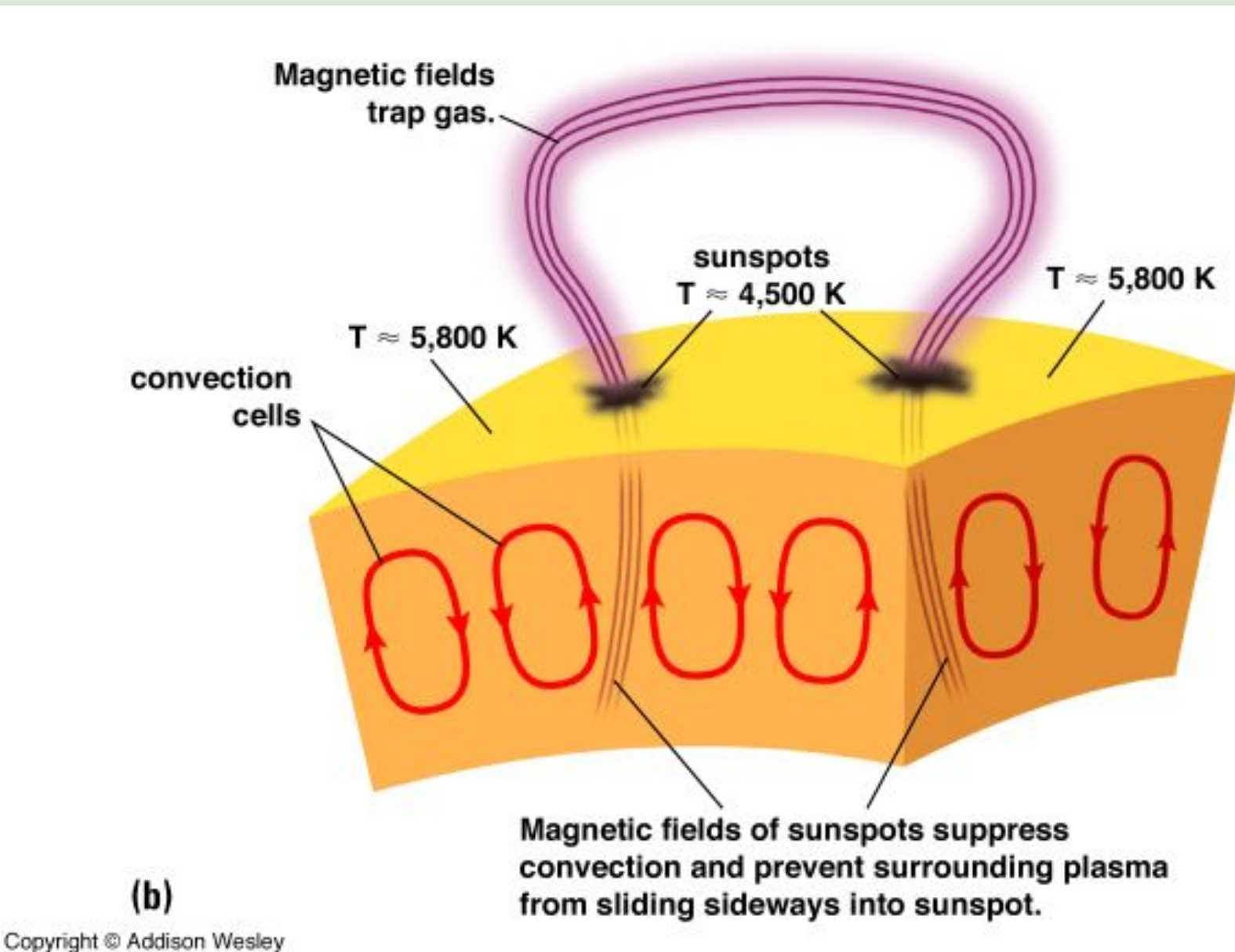


Figure 1: Sunspots form when the magnetic field forms a loop and punctures the Sun (scientificgamer.com)

Methods/Materials:

RStudio

- Conducting wavelet analysis and cross correlation function (ccf())
- Moving average (kza::kz())
- Visualizing data and results (creating graphs)
- Packages used: dplyr, tidyverse, WaveletComp, scorepeak, kza, data.table, lubridate, httr, jsonlite, maps, BBmisc

National Oceanic and Atmospheric Administration

- Data on volcanic eruption events (figure 3)

Solar Influences Data Analysis Center

- Yearly mean sunspot numbers (figure 3)

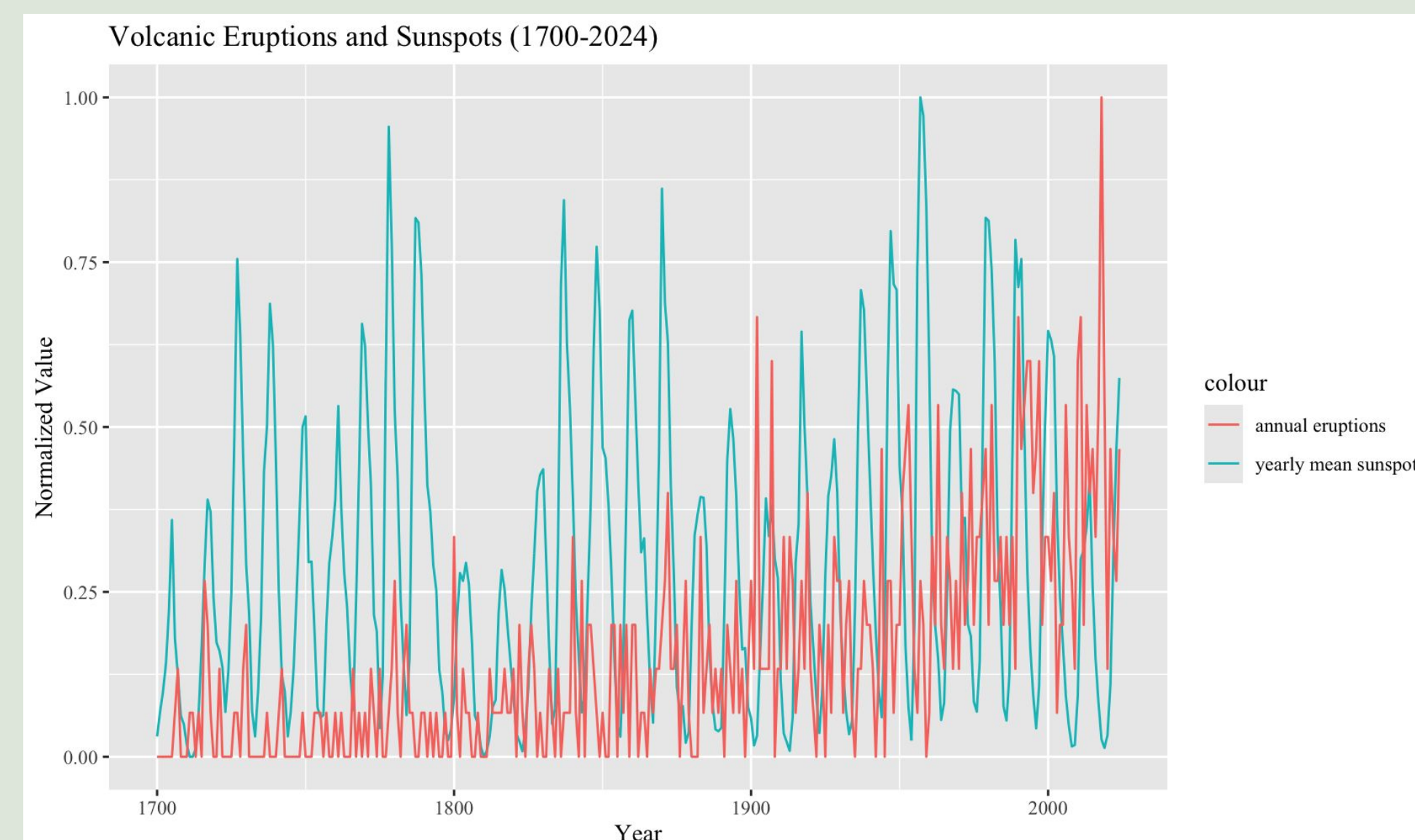


Figure 3: Normalized yearly mean sunspot numbers and total annual eruptions (1700-2024)

Results:

- Wavelet analysis of sunspot data shows a strong 10.55 year periodicity, as well as several weaker cycles (figure 4, table 1).
- Wavelet analysis of volcano data shows multiple weak periodicities (figure 5, table 2).
- Results vary slightly depending on the date range used (shown results utilize the full range of available data)
- Similar fluctuations in the periodicities appear around 200 years later in the eruption power graph.
 - This lag is supported by analysis using R's cross correlation function (an additional lag at -35 years is also observed) (figure 6)

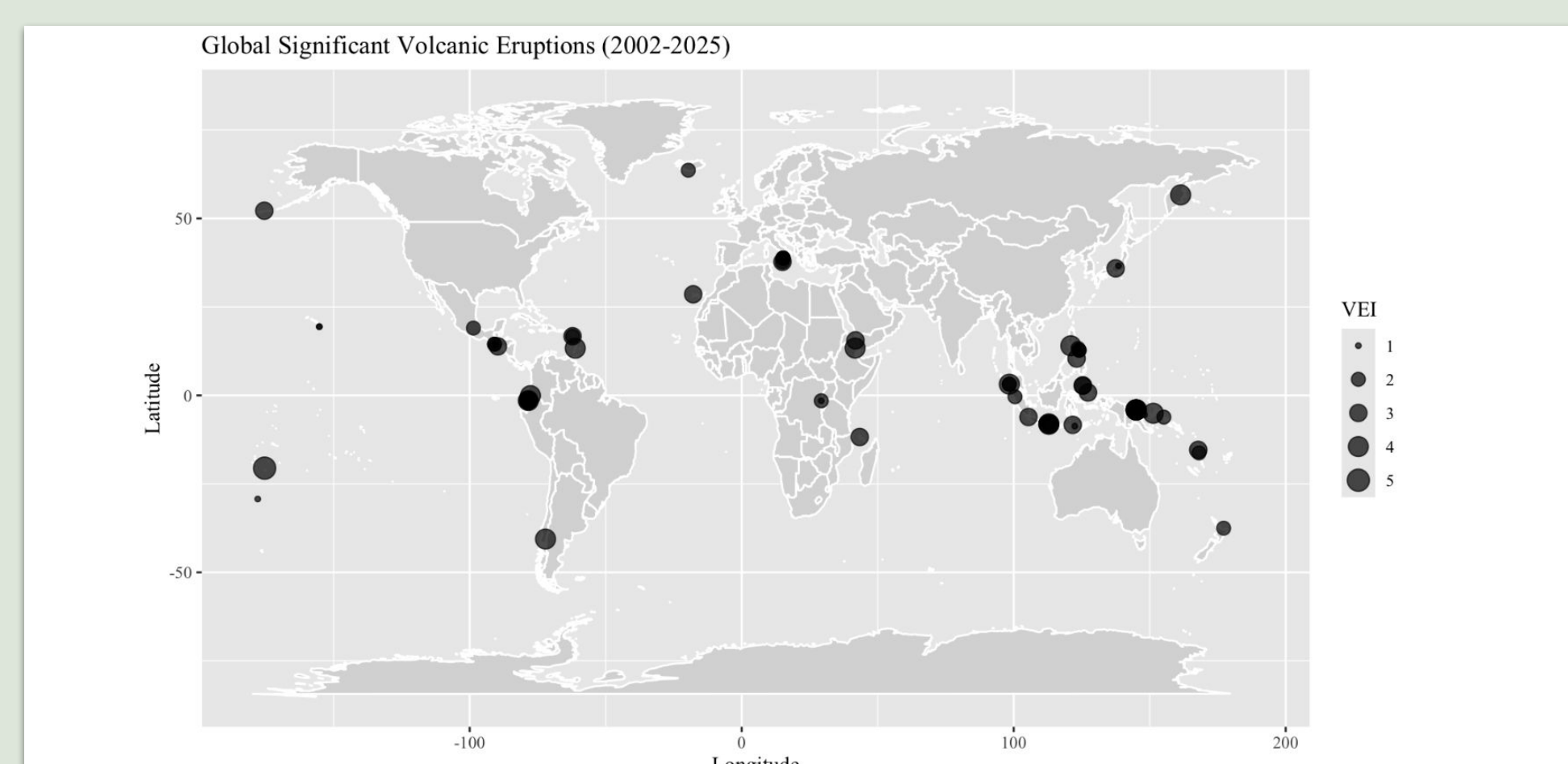


Figure 2: Circles represent volcanic eruptions that occurred between 2002 and 2025 (the size of the circle indicates the explosivity of the eruption)

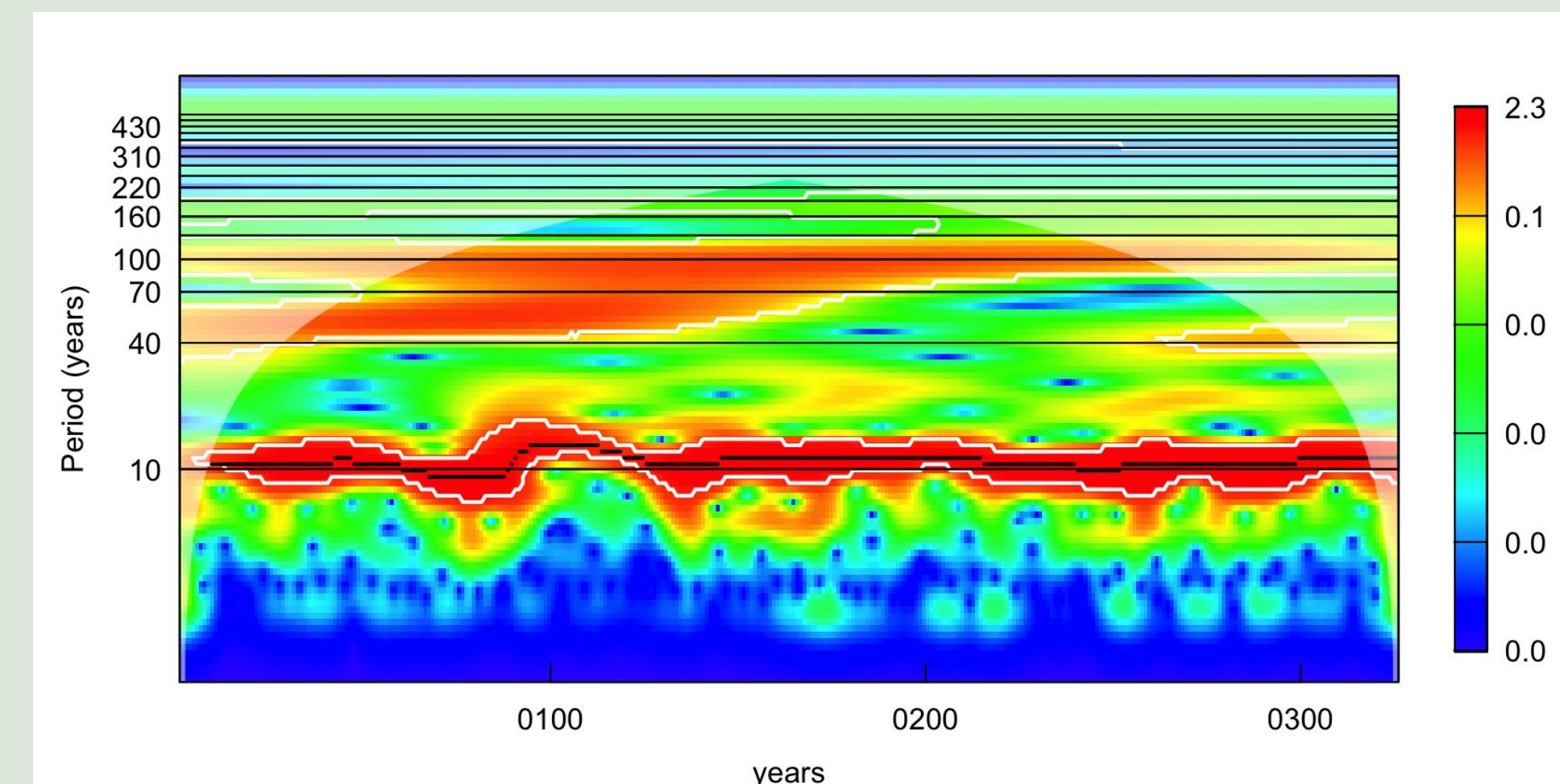


Figure 4: Wavelet power graph for a three year moving average of yearly mean sunspot numbers (1700-2024)

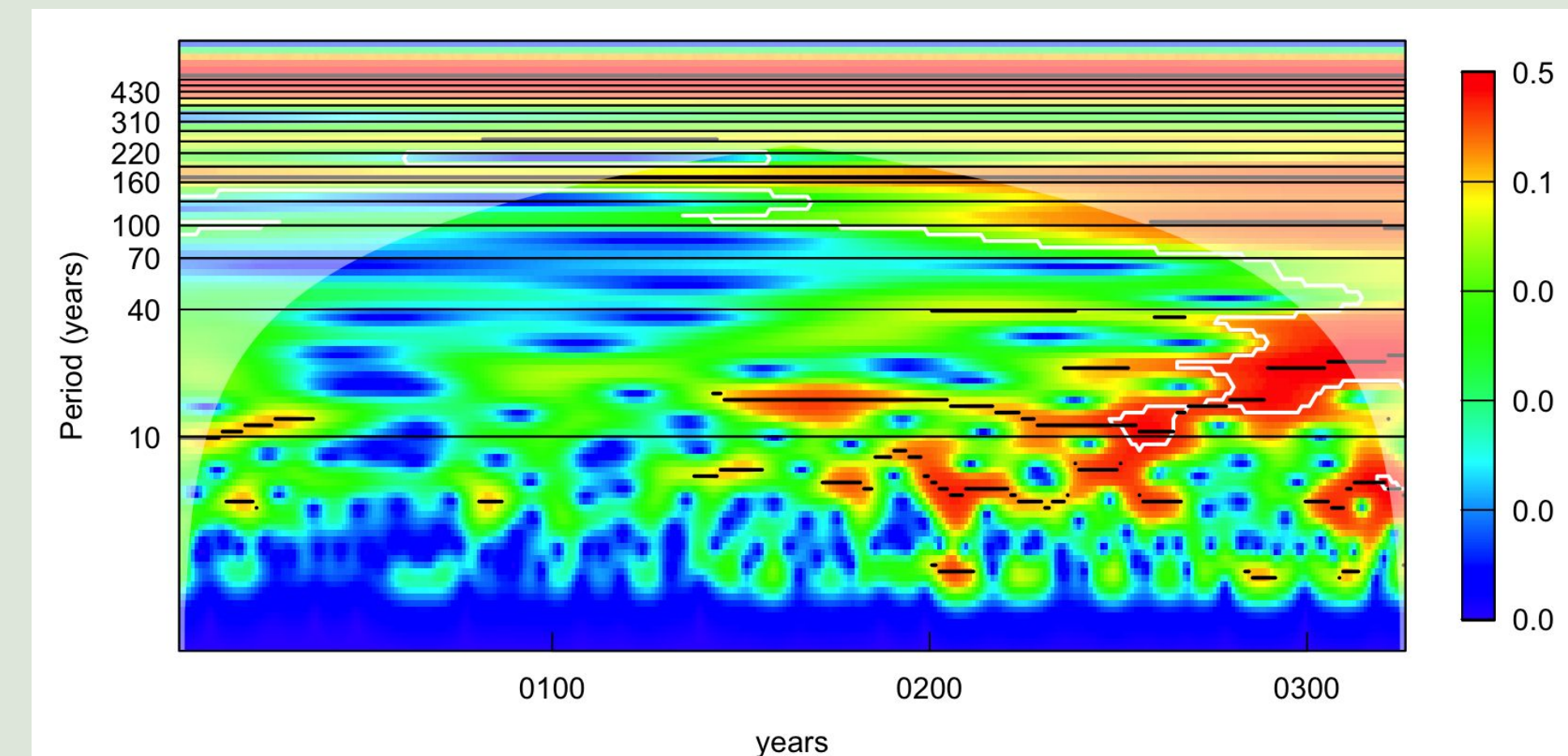


Figure 5: Wavelet power graph for a three year moving average of total annual eruptions (1700-2024)

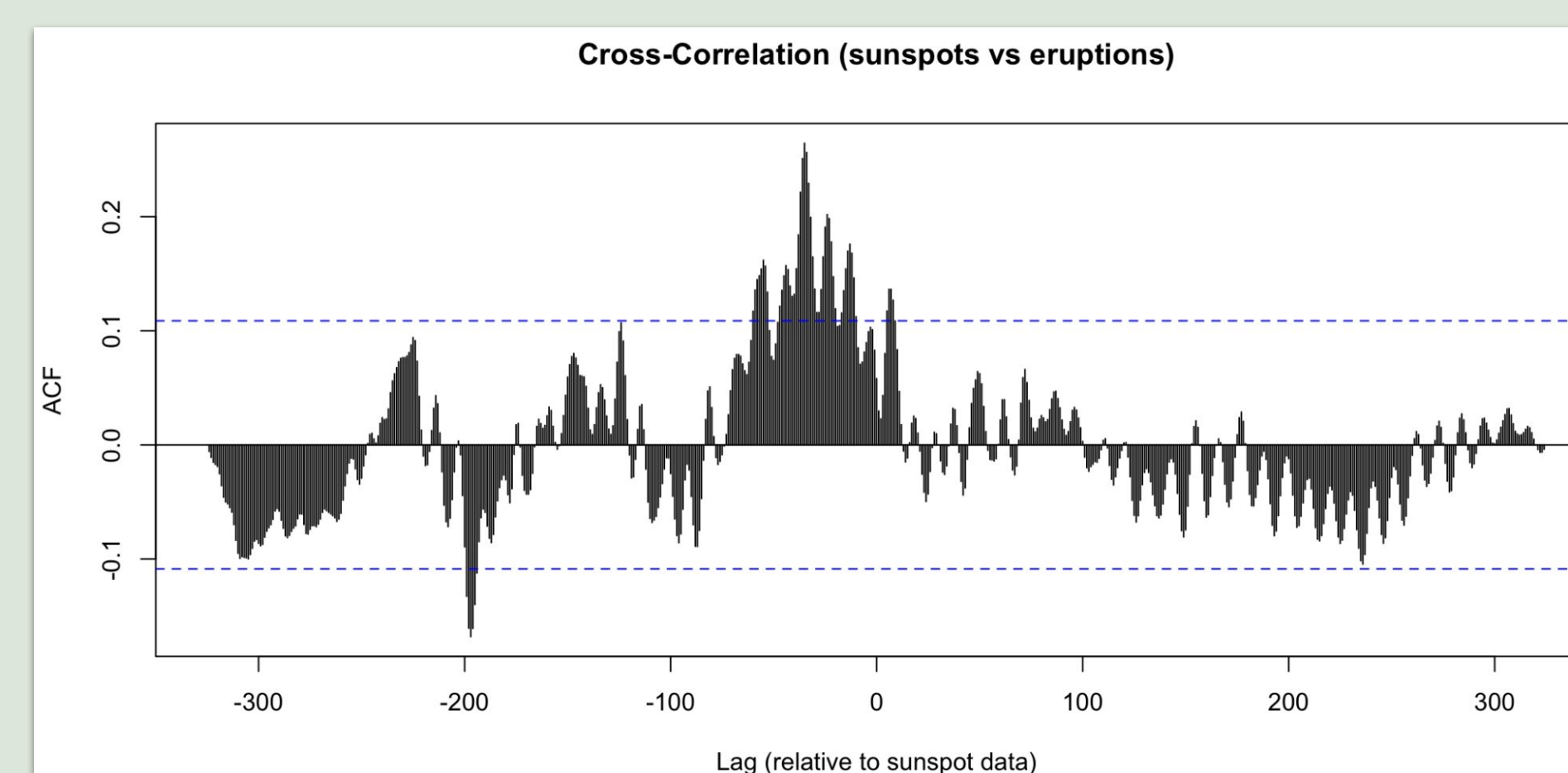


Figure 6: Cross correlation function graph comparing the three year moving averages of yearly mean sunspot numbers to total annual eruptions (1700-2024). The main significant lags are at -35 and around -200.

Discussion/Conclusions:

- There appears to be a correlation between volcanic and solar activity → why?
 - Possibly due to influence of the sun on other factors (such as the atmosphere)
- Periodicities in volcanic activity → why?
- Understanding the relationship between sunspots and volcanic eruptions → understanding the science of volcanoes → better preparation for a volcanic eruption
- Why is there a lag of 200 years to trigger a similar seasonality?

Table 1: Local peaks (periodicity and power) from the wavelet power graph in figure 3

	my.w.Power.avg.local_peaks.	my.w.Period.local_peaks.
1	0.001434001	2.297397
2	0.983333231	10.556063
3	0.031820158	21.112127
4	0.080983154	51.984153
5	0.122944966	97.005860
6	0.020807734	168.897013
7	0.002724656	256.000000
8	0.013310421	512.000000

Table 2: Local peaks (periodicity and power) from the wavelet power graph in figure 4

	my.w.Power.avg.local_peaks.	my.w.Period.local_peaks.
1	0.01839806	2.297397
2	0.05954123	5.656854
3	0.07352010	13.928809
4	0.08402118	21.112127
5	0.05279889	103.968307
6	0.09573692	168.897013
7	0.06320028	256.000000
8	0.53327582	512.000000

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Source (Google Slides):

<https://docs.google.com/presentation/d/1azT6bv-y6jg3qDb1tntjyl2hmef9wC0LXcgeb8WohwY/edit>

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